

The BioPLEASE pipeline: *retrieval-augmented planning and code-based execution for autonomous biomedical research*

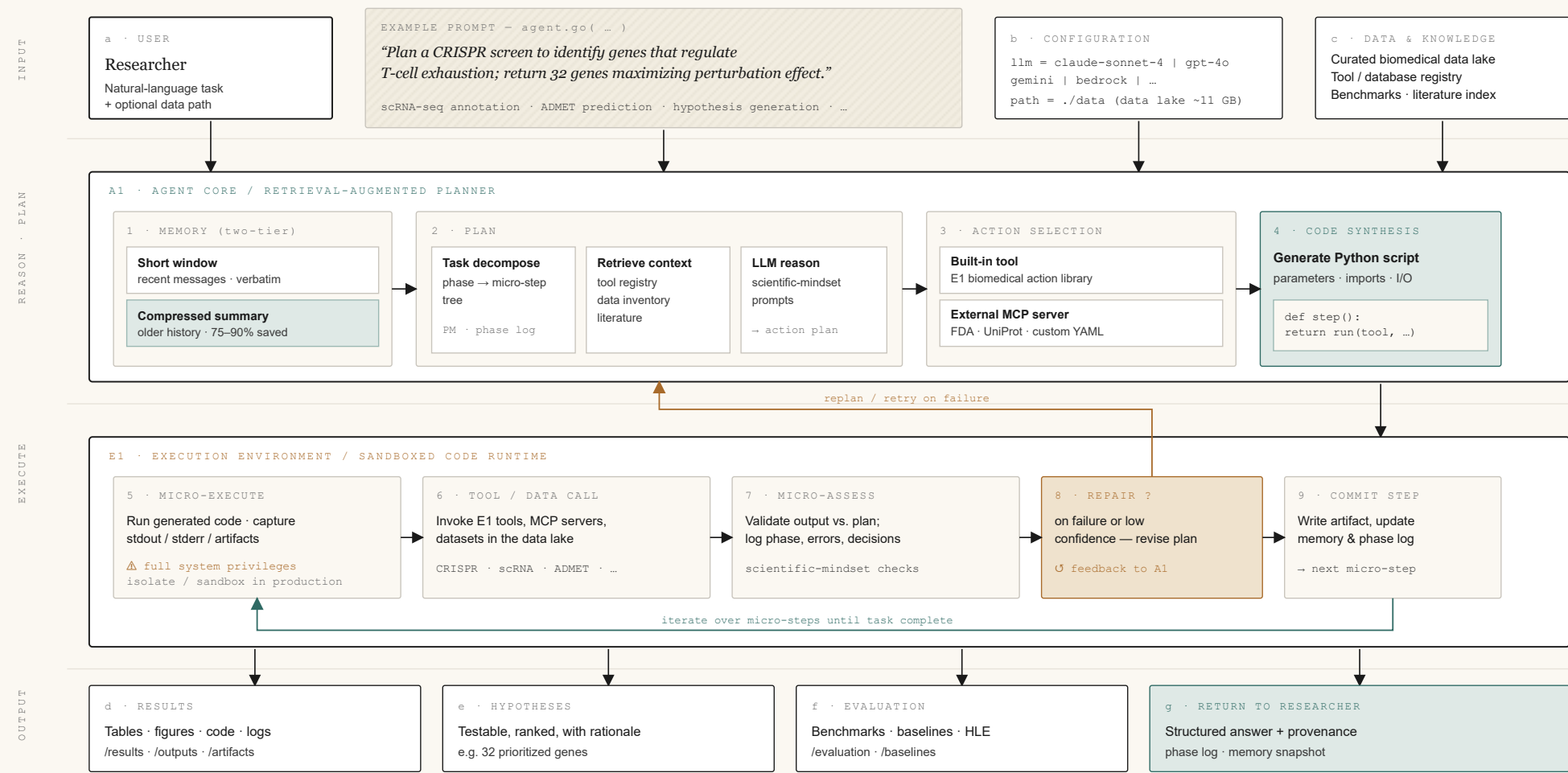


Figure 1. Overview of the BioPLEASE agent–environment pipeline. A natural-language research task (a), together with model and data-path configuration (b–c), is supplied to the **A1 agent core**. A1 maintains a two-tier memory (1) — recent messages verbatim plus a compressed summary of older history — and uses retrieval-augmented planning (2) to decompose the task into micro-steps and gather relevant tools, data and literature. For each step the planner selects an action (3) from the built-in E1 tool library or an external MCP server, and synthesizes Python code (4) that is dispatched to the **E1 execution environment**. E1 runs the code in a sandboxed runtime (5), calls biomedical tools or datasets (6), validates the result against plan and scientific-mindset checks (7), and either repairs / replans on failure (8, amber feedback) or commits the step (9). The agent iterates until the task is complete (teal loop), after which structured results, hypotheses and evaluation artifacts (d–f) are returned to the researcher with full provenance (g).